1. Agile Business Case

> ***Purpose:*** *Validates the investment, strategic alignment, and expected value delivery of the software system to secure executive funding.*

1.1 Executive Summary

ReBook is a smart, sustainable web platform that enables university students and local readers to exchange, share, and borrow academic textbooks and novels. The platform addresses the recurring financial burden of buying new books every semester and the waste of valuable books that sit idle on shelves once a semester ends. By providing a secure, centralized environment, ReBook maximizes the lifecycle of existing books instead of leaving them unused. The platform is secured with JWT-based authentication and uses Artificial Intelligence (AI) to recommend relevant textbooks to students based on their majors and reading preferences. The targeted users are university students and local readers. The estimated high-level investment required for the initial development phase (Sprints 1-6) is **$9,500**, as detailed in Section 1.4.

1.2 Problem Statement & Value Proposition

Current Operational Bottleneck / Pain Point	Proposed Software Feature / Solution	Target Metric Dynamic
Heavy financial burden on students: every semester, students face exorbitant costs when purchasing new academic books and novels, creating continuous financial pressure on them and their families.	**Sustainable economic solution:** a centralized platform that lets students acquire books at minimal cost or for free through exchanging, borrowing, and sharing.	-50% average per-semester book spending per student; 70%+ of listed books exchanged or borrowed.
Resource waste & idle books: after each semester there is no organized system on campus to recycle or reuse books, so they pile up on shelves with no educational or community benefit.	**Book lifecycle management:** digital listing and lending workflow (list, request, borrow, return) that keeps existing books in circulation and reduces paper waste.	1,000+ books listed in the first semester; 60%+ of listed books circulated at least once.
Lack of secure & organized alternatives: informal swapping (random chat groups, word of mouth) is unreliable, lacks security, and has no smart matching between what a student needs and what others have.	**Secure platform with AI matching:** JWT authentication for user data protection, plus AI algorithms that recommend relevant textbooks based on each student's major and reading preferences.	0 reported account-security breaches; 80%+ recommendation relevance rating; match time reduced from days to under 24 hours.

1.3 Strategic & KPI Alignment

- **Strategic Pillar:** Sustainability and Digital Transformation of campus resource sharing, through a circular-economy model for books and AI-enabled personalization.
- **Primary Target KPI:** Reduce students' annual book-purchasing expense by 50%, while reaching 500 active registered users and 1,000 listed books within the first semester after launch.

1.4 Cost-Benefit & ROI Analysis

Cost Category	Estimated Initial (Sprints 1-6)	Estimated Ongoing (Annual)
Cloud Architecture & Dev Infrastructure	$1,500	$1,200
Agile Engineering Team Resources	$6,000	$3,600
Third-Party API & Licensing Fees	$2,000	$1,000
Total Financial Runway	**$9,500**	**$5,800**

Cost notes: Cloud Architecture covers hosting, database, domain, and CI/CD tooling. Agile Engineering Team Resources covers development, testing, and Scrum Master / Product Owner effort across 6 sprints (ongoing: maintenance and enhancements). Third-Party API & Licensing Fees covers AI/recommendation services, email/notification services, and book-data APIs.

Expected value: if 500 active students each save about $100 per semester on books, the platform delivers roughly $50,000 in student savings per semester (about $100,000 per year) against an annual running cost of $5,800. Additional benefits include reduced paper waste, a growing reusable book inventory, and stronger campus community engagement.